

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**8th Semester of B. Tech (ME) Examination****April-May 2018****ME 407.01 Power plant engineering****Date: 30.04.2018, Monday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below.**

- (i) Mention any four requirements of good surface condensers. [02]
- (ii) State the essential requirements of draught system in thermal power plants. [02]
- (iii) Why the coal handling system is called lifeline of coal fired power plant? [02]
- (iv) _____ air is supplied separately to complete the combustion. [01]
(a) primary (b) secondary (c) tertiary (d) none of above

Q – 2.a State your opinion on effect of air leakage on performance of condenser. Also write the sources of air in condenser. [04]

Q – 2.b Answer any two questions. [10]

- (i) Describe working of forced draught system and induced draught system with necessary sketch. Which draught system is having higher heat transfer efficiency?
- (ii) Explain pollution of thermal power station and its harmful effects.
- (iii) Illustrate the working of pneumatic ash handling system. Give its advantages and disadvantages over other ash handling systems.

Q - 3 Answer any two questions. [14]

- (i) Draw the layout of combine steam and gas turbine power plant and explain its working.
- (ii) Demonstrate the construction and working of pressure compounding of steam turbine with neat sketch.

- (iii) During trial on a steam condenser, the following observations were recorded:

Condenser Vacuum = 680 mm of hg; Barometer reading = 764 mm of Hg; Mean condenser temperature = 36.2 °C; Hot well temperature = 30 °C; Condensate formed per hour = 1780 kg; Quantity of cooling water = 1250 kg/min. Cooling water inlet and outlet temperature is 20° C and 32 °C respectively. Calculate: (a) Condenser vacuum corrected to standard barometer (b) Vacuum efficiency (c) Condition of steam at inlet

of condenser (d) Condenser efficiency (e) Undercooling of condensate.

Temperature (°C)	Sat. Pressure (bar)	h_f (kJ/kg)	h_{fg} (kJ/kg)
30	0.0425	125.8	2430.5
36.2	0.06	150.9	2416.2
-	0.1119	199.7	2388.3

SECTION – II

Q - 4 Answer the question below.

- (i) Define Work ratio. If the power consumed by compressor is 100 kW in 100 kW capacity plant, Calculate Work ratio. [02]
 - (ii) State the function of control rod, coolant and reflector used in nuclear reactor. [03]
 - (iii) Unit of Air rate is _____. [01]
 - (iv) In gas turbine, net work output with reheating is _____ the net work output without reheating. [01]
- (a) less than (b) more than (c) equal to (d) unpredictable

Q – 5.a Briefly explain the factors to be considered for site selection of modern thermal power plant. [06]

OR

Q – 5.a Suggest the boiler for steam generation at 225 bar pressure. Explain its working with necessary diagram. [06]

Q – 5.b What is tariff? What is the significance of two part tariff and three part tariff? [04]

Q – 5.c Compare nuclear fusion and fission process. [04]

OR

Q – 5.c Draw the neat sketch of BWR and explain its working. [04]

Q – 6.a Draw h-s diagram and layout of open cycle gas turbine with reheating and intercooling and derive the equation of efficiency of ideal and actual cycle. [06]

Q – 6.b A gas turbine plant receives air at a pressure of 1 bar and 290 K. The air is then compressed in a rotary compressor to a pressure of 4 bar and then heated to a temperature of 840 K. The isentropic efficiencies of compressor and turbine are 82% and 86% respectively. Calculate: (1) Net Work done (2) Work ratio (3) Heat supplied (4) overall thermal efficiency. Also calculate overall thermal efficiency, if heat exchanger is provided with 70% effectiveness. Neglect the pressure drop during entire heat transfer process. Draw h-s diagram for the same. [08]

OR

Q – 6.a State any four practical application of Gas Turbine. Draw p-V and T-S diagram and [07]
derive the equation of efficiency ideal brayton cycle.

Q – 6.b The following observations were made on a boiler plant during one hour test: [07]

Steam pressure = 20 bar; Steam temperature = 260 °C; Steam generated = 37,500 kg;
Temperature of water entering the economiser = 15 °C; Temperature of water leaving
the economiser = 90 °C; Fuel used = 4400 kg; Calorific value of fuel = 30,000 kJ/kg.

Calculate: (1) The equivalent evaporation per kg of fuel (2) Thermal efficiency of the
plant and (3) the percentage heat energy of the fuel energy utilized by the economiser.

At $p = 20$ bar, Saturation temperature (T_{sat}) = 212.4 °C and enthalpy of dry and
saturated vapour (h_g) = 2797.2 kJ/kg.
